Restricted Exhaustive Search for Frequency Band Selection in Motor Imagery Classification

Paul Bustios, *Student Member, IEEE*, João Luís Rosa, *Senior Member, IEEE*
Institute of Mathematical and Computer Sciences
University of Sao Paulo
pbustios@usp.br, joaoluis@icmc.usp.br

***Abstract*—Motor imagery is a mental process that produces modulations in the amplitude of ongoing electroencephalogram signals. The patterns present in these modulations can be used to classify this mental process, but the identification of these patterns is not a trivial task, because they are present in frequency bands that are specific for each person. In this paper, we introduce a novel method to select these subject-specific frequency bands in a new configuration for the Filter Bank Common Spatial Pattern approach. Our method uses an exhaustive search to find the best subset of frequency bands containing the most discriminative patterns within a search space restricted to a fixed size for this subset. The size is determined using cross-validation and the Sequential Forward Floating Selection method. As we demonstrate with experiments on the data set 2b of the BCI Competition IV, our method is more accurate than current approaches evaluated on this data set.**

I. Introduction

An electroencephalogram (EEG) is a record of electrical activity in the brain measured using electrodes attached to the scalp and is composed of several signals, usually one for each electrode. EEGs are widely used for brain research and brain-computer interfaces because of their high time resolution measurements, non-invasive recording procedure and relatively inexpensive recording equipment compared with other devices. Within EEG analysis topic, classification of motor imagery EEG signals is an issue that has received considerable attention due to its importance in helping people with physical disabilities [1], [2], [3], [4] and in other applications like virtual world navigation [5] and entertainment [6], [7].

Motor imagery (MI) is a mental process that consists in rehearsing a movement mentally without any kind of muscular activity as a result [4]. This mental process yields a series of phenomena that cause changes in the amplitude of the ongoing EEG signals within the sensorimotor (8-30 Hz) rhythms [8]. These phenomena are called Event-Related Desynchronization (ERD), which represents an amplitude decrease [9], and Event-Related Synchronization (ERS), which represents an amplitude increase [10], but the identification and classification of these events is not a trivial task, because the brain activity is different among people, which means that ERD/ERS are frequency-band specific for each person [11], [12].

Several methods have been proposed to address that issue, being Filter Bank Common Spatial Pattern (FBCSP) [13], one of the most successfull methods in classification of motor imagery EEG signals and winner of the BCI Competition IV for the data sets 2a and 2b [14]. FBCSP extracts a set of features applying the Common Spatial Pattern (CSP) method [13] on EEG signals band-pass filtered into multiple frequency bands. Then, features from some frequency bands are selected for classification. Any method can be used to select the features and make the classification.

An exhaustive search to find the frequency bands containing the most discriminative features can be the most effective method, but it is not popular because it is time consuming. However, if we knew how many frequency bands are enough to obtain a good performance in classification, the exhaustive search would be reduced to finding the best combination of only that number of frequency bands from all available in the filter bank.

In this paper, we introduce a novel method to address this observation. First, we evaluate how well the features extracted from a subset of frequency bands will perform in classification. The frequency bands in a subset are selected using the Sequential Forward Floating Selection method [15] with the average accuracy of a classifier in cross-validation as criterion function to measure the significance of the features. We start searching for a subset of size one and stop the search when the subset of size $n+1$ does not offer a significant difference in the results of cross-validation with respect to the previous subset of size n. Then, it is performed an exhaustive search in a reduced search space restricted to finding the best combination of n frequency bands from all available in the filter bank.

Our two-step frequency band selection method is part of a new configuration for the FBCSP method. We evaluated the performance of this new configuration on the data set 2b of the BCI Competition IV [16]. As we will demonstrate, our method can improve not only the accuracy of classification of motor imagery EEG signals, but also the efficiency using features from only two frequency bands.

The rest of this paper is organized as follows. In Section II, we briefly describe the FBCSP method before explaining our proposed method in Section III. In Section IV, we provide an evaluation of our method and in Section V, we offer conclusions and directions for future work.

II. Filter Bank Common Spatial Pattern

The FBCSP method is composed of four progressive stages as illustrated in figure 1. In the first stage, EEG signals are

978-1-5090-6182-2/17/$31.00 ©2017 IEEE

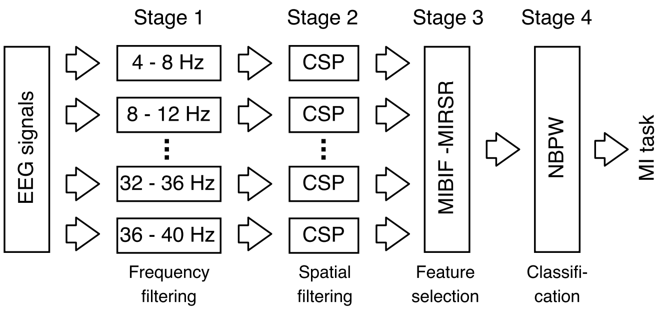

Fig. 1. Architecture of FBCSP method, adapted from [13].

filtered using a filter bank composed of nine band-pass filters in the frequency bands of 4-8, 8-12, 12-16, ..., 36-40 Hz. The second stage employs the CSP method for feature extraction. The third and fourth stages perform feature selection and classification respectively. Any method can be used in the last two stages, but for the BCI Competition IV, FBCSP was configured with the Mutual Information-based Best Individual Feature (MIBIF) and the Mutual Information-based Rough Set Reduction (MIRSR) algorithms for feature selection and a Naive Bayesian classifier using the Parzen-window (NBPW) algorithm for classsification [13].

III. Our method

Our method for frequency band selection is part of a new configuration for the FBCSP method. This new configuration has two phases, one to select the frequency bands containing the discriminative features and the other to classify the MI tasks.

The frequency band selection phase is composed of three stages as shown in figure 2. In the first stage, each segment of EEG, hereinafter called epoch, used for training is band-pass filtered into ten different frequency bands. Then, in the second stage, a set of features, hereinafter called CSP features, is extracted employing the CSP method. Finally, in the third stage, we use the Sequential Forward Floating Selection (SFFS) method and a Restricted Exhaustive Search (RES) within a reduced search space to determine which frequency bands keep discriminative patterns for classification.

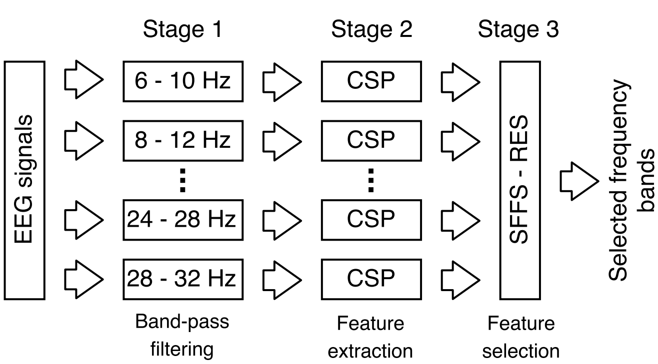

Fig. 2. Architecture for the frequency band selection phase of our method.

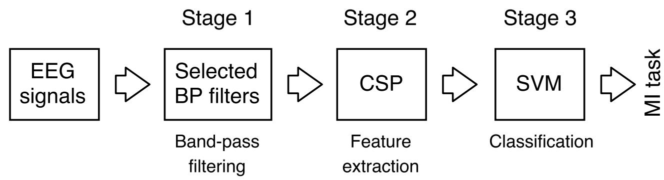

Fig. 3. Architecture for the classification phase of our method.

For the classification phase, our method is also composed of three stages as illustrated in figure 3. But with the difference that now, the epochs are just band-pass filtered into the frequency bands selected in the previous phase; and in the last stage, a Support Vector Machine (SVM) classifier [17] is introduced to determine the class of the CSP features.

A. Band-pass filtering

Since we are interested in the sensorimotor rhythms, which are around of 8-30 Hz [1], [8], [18], [11], [19], we use a filter bank that covers that frequency range, composed of 10 Chebyshev Type II band-pass filters, each of them with a bandwidth equal to 4 Hz and an overlap of 2 Hz with the next filter, namely 8-12, 10-14, ..., 24-28, 26-30 Hz as in [20]. This configuration is employed only in the frequency band selection phase. For the classification phase, the epochs are band-pass filtered into the selected frequency bands.

B. Feature extraction

As in FBCSP, our method employs the CSP procedure to find a spatial filter for each frequency band. CSP was first described in [21] to extract spatial patterns of EEGs from normal subjects and patients with neurological disorders. The CSP mathematical procedure for a two-class problem is described below.

First, calculate the covariance matrices for each class $k \in \{a, b\}$ of the training epochs band-pass filtered into one of the ten frequency bands:

$$R_k = \frac{1}{n_k} \sum_{i=1}^{n_k} X_{k,i} X_{k,i}^T \tag{1}$$

where:

- $X_{k,i} \in \mathbb{R}^{c \times t}$ is the i-th epoch of the class k, composed of t samples from c channels.
- n_k is the number of epochs of class k.
- $R_k \in \mathbb{R}^{c \times c}$ is the covariance matrix of the band-pass filtered epochs of the class k.

Then, with the covariance matrices found using equation 1, solve the generalized eigenvalue problem in the matrix form:

$$R_a W = (R_a + R_b) W \Lambda \tag{2}$$

where:

- $W \in \mathbb{R}^{c \times c}$ (with c = number of channels) contains the generalized eigenvectors $w_{j=\{1,\ldots,c\}}$ of R_a and $R_a + R_b$.

- $\Lambda \in \mathbb{R}^{c \times c}$ is a diagonal matrix which contains the generalized eigenvalues $\lambda_{j=\{1,...,c\}}$, corresponding to the eigenvectors w_j, of R_a and $R_a + R_b$.

Finally, if the eigenvalues λ_j are arranged in a non-increasing order, i.e. $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_c$, and their corresponding eigenvectors w_j share the same arrangement, then, the first m eigenvectors in W will account for high variance for class a and low variance for class b. Conversely, the last m eigenvectors will account for low variance for class a and high variance for class b [22]. Thus, W can be considered a set of spatial filters.

Since the data set used in this work has three EEG channels, W will be composed of three eigenvectors. Hence the two obvious options to compose our matrix of spatial filters are: to use all the three eigenvectors as our spatial filters or just the two eigenvectors that account for the highest/lowest variance for each class of MI. In this work, we chose the latter option. Thus, we used the first and the last eigenvectors of W to compose our set of spatial filters W_{CSP}.

To extract the features of an epoch X, we apply the following equation:

$$f = \log\left(\frac{diag(W_{CSP}^T X X^T W_{CSP})}{t}\right) \tag{3}$$

where:

- $W_{CSP} \in \mathbb{R}^{c \times m}$ is a set of m spatial filters (with c = number of channels and m = number of selected eigenvectors).
- $diag(\cdot)$ returns the main diagonal elements of a squared matrix.
- t is the number of samples in the epoch X.
- $f \in \mathbb{R}^2$ is the CSP feature.

Thus, applying the CSP mathematical procedure (equation 1, 2 and 3) in the frequency band selection phase, we can obtain a feature vector $F = [f_1, f_2, ..., f_{10}]$ from a training epoch X, where each CSP feature f_i corresponds to the epoch band-pass filtered in one of the ten frequency bands in the filter bank.

C. Feature selection (Frequency band selection)

In this stage, we select the most discriminative CSP features in order to determine which frequency bands contain them. Also, this selection will make classification more efficient and interpretable. First we explain the methods that compose our frequency band selection method.

1) Sequential Forward Floating Selection: The SFFS method is a feature selection method that was proposed to avoid the so-called "nesting effect" in the sequential backward selection and sequential forward selection methods [15]. Though SFFS does not guarantee the best subset of features, its performance is good compared to other feature selection methods. The SFFS method is described below.

Let Y be the set of all the CSP features, S an empty set that will contain the selected CSP features and $J(\cdot)$ our criterion function to measure the discriminative significance of a CSP feature:

- Step 1: using $J(\cdot)$, search for the most significant CSP feature in Y. Then add it to S.
- Step 2: search for the next CSP feature f in $Y - S$ that maximizes $J(S \cup f)$. Then add f to S.
- Step 3: again, search for the next CSP feature f in $Y - S$ that maximizes $J(S \cup f)$. Then add f to S.
- Step 4: find the least significant CSP feature $f_i \in S$. If the latest added CSP feature f is the least significant, i.e. $J(S - f) \geq J(S - f_i)$, $\forall f_i \in \{S - f\}$, then return to step 3, else exclude f_i from S. If the size of S is two, return to step 3, else go to step 5.
- Step 5: find the least significant CSP feature $f_i \in S$. f_i can be removed from S if a new combination of later selected CSP features achieves better results. If the size of S is two, return to step 3, else repeat step 5.

In this work, our criterion function $J(\cdot)$ is the average accuracy of a Support Vector Machine (SVM) classifier over all repetitions of a 10 × 10-fold cross-validation procedure.

2) Restricted Exhaustive Search: Since the SFFS method does not guarantee that we will obtain the best n CSP features, we employ a restricted exhaustive search (RES) to find them. The restriction consist in finding only the most significative n-combination of all available CSP features. Thus, this restriction will reduce the search space from 2^m to $m!/(n!(m-n)!)$ possible combinations, where m is the number of band-pass filters. Since each CSP feature is related to a frequency band, the selected CSP features will determine which frequency bands are relevant to classify the MI tasks.

The significance of a CSP feature is measured with the same criterion function used in the SFFS method.

With the two methods explained above, we select the frequency bands following a two-step procedure as shown in the next algorithm:

1: $Y \leftarrow$ Set of CSP features of all subjects
2: $n \leftarrow 1$
3: $S_{n+1} \leftarrow \textit{SFFS}(Y, n)$
4: **repeat**
5: $\quad S_n \leftarrow S_{n+1}$
6: $\quad n \leftarrow n + 1$
7: $\quad S_{n+1} \leftarrow \textit{SFFS}(Y, n)$
8: **until** $CV(S_{n+1}) \approx CV(S_n)$
9: $n \leftarrow n - 1$
10: $S \leftarrow \textit{RES}(Y, n)$
11: **return** S

- In the first step (lines 1-8), we employ the SFFS method to determine, for all subjects in parallel, how many frequency bands are enough to obtain discriminative patterns to classify the MI tasks. We start searching for a subset of size one (line 3) and stop the search when the subset of size $n + 1$ does not offer a significant difference in the results of cross-validation (CV) with respect to the previous subset of size n (line 8). We use a paired t-test to verify the significance of the results and

set our significance level α at 0.05. The optimal number of frequency bands will be denoted by n.
- In the second step (lines 9-11), it is performed an exhaustive search restricted to finding, for all subjects in parallel, the best combination of n frequency bands from all available in the filter bank. Then, the selected frequency bands for each subject are returned in S.

D. Classification

The features extracted from the selected subset of frequency bands are classified using a Support Vector Machine (SVM) model [17]. This supervised learning model maps the features to a very high-dimensional space and uses only a subset of them, called support vectors, to create a linear decision function. We choose this discriminative classifier because it has high generalization ability, it comes with theoretical guarantees about the performance and it is not affected by local minima. We have configured our SVM model with a paramater $C = 0.04$ and a linear kernel, which is less prone to overfitting than non-linear kernels.

IV. Experiments and Results

To ensure that our experiments are easily reproducible, we make our code available in a public repository [23]. In addition, this repository contains more data about the results.

We have evaluated our method on the data set 2b of the BCI Competition IV [16]. This data set includes EEGs composed of three bipolar recordings (C3, Cz, and C4) from nine subjects with a sampling frequency of 250 Hz. For each subject, there are five recording sessions: three for training (01T, 02T and 03T) and two for evaluation (04E and 05E). Each session contains several trials of two classes of MI, namely MI of left hand and MI of right hand. Sessions 01T and 02T were recorded without feedback, and sessions 03T, 04E and 05E were recorded with feedback, i.e. the subjects knew if they were doing well (green happy smiley) or not (red sad smiley). Figure 4 illustrates the recording protocol for each trial of each type of session.

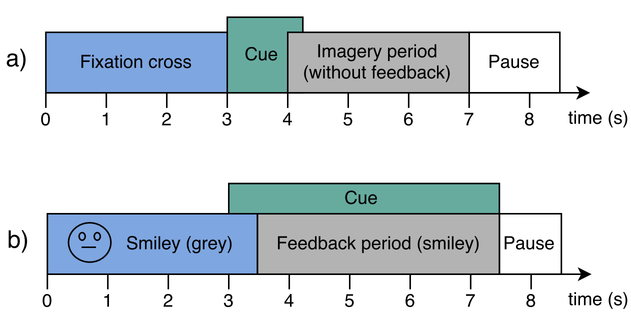

Fig. 4. Recording protocols. (a) Trials in sessions 01T and 02T were recorded without feedback. (b) For the trials in sessions 03T, 04E and 05E, the feedback started at second 3.5 and ended at second 7.5. Adapted from [16].

For the trials of the sessions without feedback, a cue at second 3 with a duration of 1.25 seconds was presented to indicate what MI task had to be performed. These type of trials have a duration of 7 seconds. For the trials of the sessions with feedback, a cue was presented from second 3 to 7.5. The duration of these trials is 7.5 seconds. More details about this data set can be found in [5], [16].

TABLE I
Estimated accuracy employing 10×10-fold cross-validation and features from 1 to 5 frequency bands selected with SFFS. The selected frequency bands for each subject can be found on the public repository [23]

	CV Accuracy (%)				
Subject	1 Band	2 Bands	3 Bands	4 Bands	5 Bands
1	79.68	82.29	82.57	82.04	82.11
2	58.80	61.62	63.15	64.47	64.15
3	60.25	60.72	63.25	64.10	63.58
4	99.69	100.0	100.0	100.0	100.0
5	83.75	88.63	89.31	90.94	90.75
6	79.44	83.75	84.12	84.81	84.69
7	89.81	90.44	90.81	91.06	92.81
8	90.19	90.94	91.00	90.75	90.69
9	85.06	85.19	84.75	85.81	85.69
Average	80.74	**82.62**	83.22	83.78	83.83

We have performed the following experimental procedure. To classify a sample x_t, it was taken 499 samples before it, namely $x_{t-499}, x_{t-498}, ..., x_{t-1}$. Thus, for both the frequency band selection and the classification phases, we used epochs of 2 seconds (500 samples). For the frequency band selection, we used epochs starting at second 3.5 from all trials of: sessions 01T and 03T for subject 1; sessions 01T, 02T and 03T for subjects 2 and 3; session 03T for subjects 4, 5, 6, 7, 8 and 9 as in [14]. We applied cross-validation to corroborate that the trials of these sessions provide better results than other possible combinations of sessions. These epochs were also used to find out the spatial filters and to train the SVM classifier in the classification phase.

The first step is to find the optimal number of frequency bands to be used in the classification phase. As shown in Table I, our frequency band selection method found that two is an optimal number. There is a significant improvement between the results of cross-validation using a subset of one and two frequency bands ($p = 0.015$). On the other hand, the results of cross-validation using a subset of three frequency bands do not show a significant improvement with respect to the results of using a subset of two frequency bands ($p = 0.083$). Similarly, using four or five frequency bands, the improvement in the results of cross-validation is not significant.

Once determined the optimal number of frequency bands, the next step is to use the RES method to find the best combination of two frequency bands for each subject. The results of this search are shown in Table II. As we can see, there is an improvement in the results of cross-validation using the frequency bands found by the RES method. Only these frequency bands will be used to filter the epochs in the classification phase.

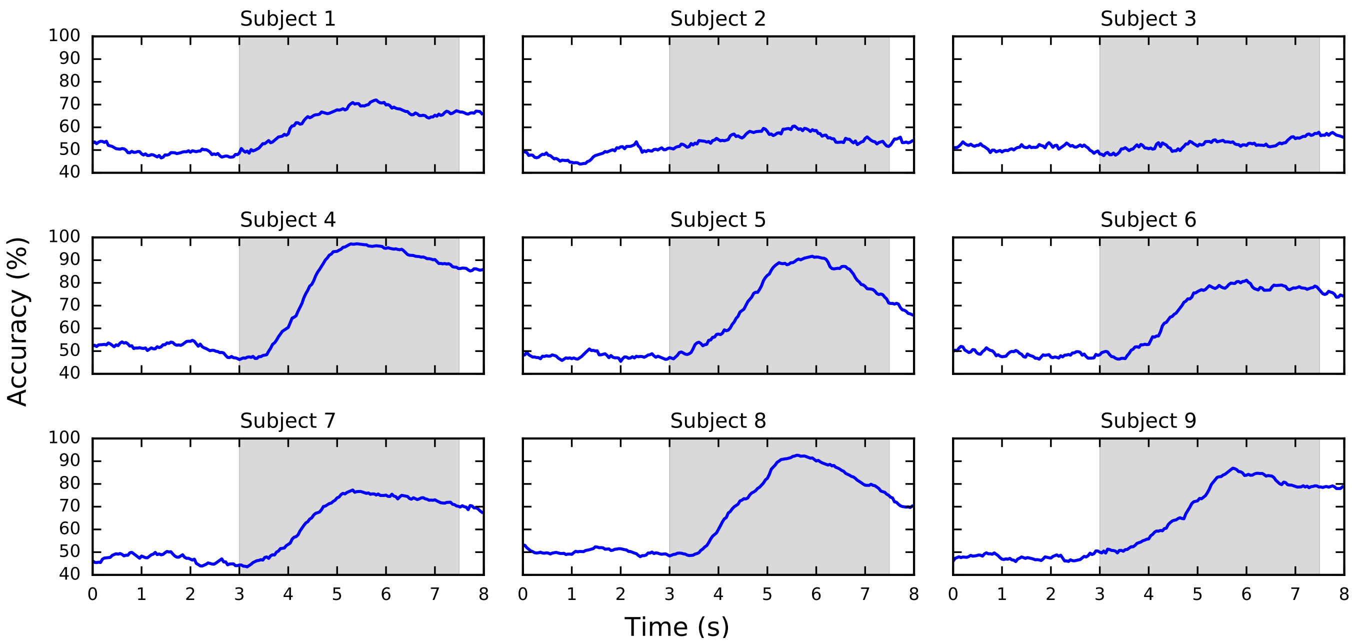

Fig. 5. Time course of the accuracy scores of our method for the trials of the evaluation sessions. The shaded parts indicate the time when the cue was presented to ask the subjects to perform a specific MI task.

TABLE II
SELECTED FREQUENCY BANDS USING RESTRICTED EXHAUSTIVE SEARCH.

Subject	Selected frequency bands (Hz)	CV Accuracy (%)
1	10 - 14, 20 - 24	82.29
2	10 - 14, 20 - 24	61.62
3	8 - 12, 18 - 22	61.72
4	8 - 12, 10 - 14	100.0
5	22 - 26, 26 - 30	88.63
6	10 - 14, 12 - 16	83.75
7	12 - 16, 18 - 22	91.50
8	8 - 12, 10 - 14	90.94
9	18 - 22, 22 - 26	85.19
Average		**82.85**

TABLE III
CLASSIFICATION RESULTS OF THE EVALUATION DATA OF THE BCI COMPETITION IV DATA SET 2B WITH THE KAPPA COEFFICIENT AS METRIC.

	Kappa values			
Subject	aFBCSP [24]	FBCSP [14]	HCRF [25]	Our method
1	0.406	0.400	**0.560**	0.462
2	0.179	0.207	0.240	**0.343**
3	0.150	0.219	0.180	**0.263**
4	**0.963**	0.950	0.950	0.956
5	0.863	0.856	**0.930**	0.856
6	0.606	0.613	**0.720**	0.656
7	0.575	0.550	0.510	**0.631**
8	0.856	0.850	0.800	**0.863**
9	0.738	0.744	0.710	**0.756**
Average	0.593	0.599	0.622	**0.643**

As required by the BCI Competition IV, we provide a continuos classification output for each sample. Figure 5 shows the classification accuracy for each sample of the trials of the evaluation sessions. The classification accuracy was not good for subjects 2 and 3, because they had difficulty in performing the MI tasks [5].

We report the maximum kappa values in Table III and maximum accuracy scores in Table IV achieved throughout the course of time of the trials of the evaluation sessions for each subject. The results are compared to other results of approaches that were also evaluated on the data set 2b of the BCI Competition IV, showing that, on average, our method is more accurate. In addition, our method uses only two frequency bands for classification.

[1]We present the average of the classification accuracies of the evaluation sessions (04E and 05E) presented in [27] in order to compare the results.

V. CONCLUSION AND FUTURE WORK

We have introduced a novel frequency band selection method for accurate classification of motor imagery EEG signals. We have demonstrated on a real-world and publicly available data set that our approach is more accurate than approaches evaluated on the same data set. In particular, we have shown that our method achieved better results using few band-pass filters. In future work, we plan to extend our method to a multi-class problem. Finally, we have made our code publicly available [23] to allow others to confirm, extend and use our work.

TABLE IV
CLASSIFICATION RESULTS OF THE EVALUATION DATA OF THE BCI COMPETITION IV DATA SET 2B WITH ACCURACY SCORE AS METRIC.

Subject	Accuracy score (%)		
	UB Eval [26]	EMD based filtering [27] [1]	Our method
1	70.31	62.80	**73.13**
2	61.25	67.05	**67.14**
3	59.69	**98.70**	63.13
4	97.19	88.40	**97.81**
5	90.94	**96.25**	92.81
6	**85.63**	75.25	82.81
7	78.44	72.15	**81.56**
8	88.75	87.80	**93.13**
9	78.44	85.25	**87.81**
Average	78.96	81.52	**82.15**

ACKNOWLEDGMENT

P. Bustios would like to thank the Brazilian funding agencies Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) for the financial support. J. L. G. Rosa is grateful to the São Paulo Research Foundation FAPESP, process 2016/02555-8, for the financial support.

REFERENCES

[1] G. Pfurtscheller, C. Guger, G. Müller, G. Krausz, and C. Neuper, “Brain oscillations control hand orthosis in a tetraplegic,” *Neuroscience Letters*, vol. 292, no. 3, pp. 211–214, oct 2000.

[2] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, “Braincomputer interfaces for communication and control,” *Clinical Neurophysiology*, vol. 113, no. 6, pp. 767–791, jun 2002.

[3] N. Birbaumer and L. G. Cohen, “Brain-computer interfaces: communication and restoration of movement in paralysis,” *The Journal of Physiology*, vol. 579, no. 3, pp. 621–636, mar 2007.

[4] T. Mulder, “Motor imagery and action observation: Cognitive tools for rehabilitation,” *Journal of Neural Transmission*, vol. 114, no. 10, pp. 1265–1278, 2007.

[5] R. Leeb, F. Lee, C. Keinrath, R. Scherer, H. Bischof, and G. Pfurtscheller, “Brain-computer communication: Motivation, aim, and impact of exploring a virtual apartment,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 15, no. 4, pp. 473–482, dec 2007.

[6] A. Nijholt, B. Reuderink, and D. O. Bos, “Turning shortcomings into challenges: Brain-computer interfaces for games,” *Entertainment Computing*, vol. 1, no. 2, pp. 85–94, 2009.

[7] L. Bonnet, F. Lotte, and A. Lecuyer, “Two Brains, One Game: Design and Evaluation of a Multiuser BCI Video Game Based on Motor Imagery,” *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 5, no. 2, pp. 185–198, jun 2013.

[8] G. Pfurtscheller and C. Neuper, “Motor imagery activates primary sensorimotor area in humans,” *Neuroscience Letters*, vol. 239, no. 2, pp. 65–68, 1997.

[9] G. Pfurtscheller and A. Aranibar, “Event-related cortical desynchronization detected by power measurements of scalp EEG,” *Electroencephalography and Clinical Neurophysiology*, vol. 42, no. 6, pp. 817–826, 1977.

[10] G. Pfurtscheller, “Event-related synchronization (ERS): an electrophysiological correlate of cortical areas at rest,” *Electroencephalography and Clinical Neurophysiology*, vol. 83, no. 1, pp. 62–69, 1992.

[11] G. Pfurtscheller and F. Lopes da Silva, “Event-related EEG/MEG synchronization and desynchronization: basic principles,” *Clinical Neurophysiology*, vol. 110, no. 11, pp. 1842–1857, nov 1999.

[12] G. Pfurtscheller, “Functional brain imaging based on ERD/ERS,” *Vision Research*, vol. 41, no. 10, pp. 1257–1260, 2001.

[13] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, “Filter Bank Common Spatial Pattern (FBCSP) in Brain-Computer Interface,” in *International Joint Conference on Neural Networks (IJCNN)*, no. July 2016. IEEE, jun 2008, pp. 2390–2397.

[14] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, “Filter Bank Common Spatial Pattern Algorithm on BCI Competition IV Datasets 2a and 2b.” *Frontiers in neuroscience*, vol. 6, p. 39, jan 2012.

[15] P. Pudil, J. Novovičová, and J. Kittler, “Floating search methods in feature selection,” *Pattern Recognition Letters*, vol. 15, no. 11, pp. 1119–1125, nov 1994.

[16] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, and B. Blankertz, “Review of the BCI Competition IV.” *Frontiers in neuroscience*, vol. 6, p. 55, jan 2012.

[17] C. Cortes and V. Vapnik, “Support-Vector Networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[18] G. Pfurtscheller, C. Neuper, D. Flotzinger, and M. Pregenzer, “EEG-based discrimination between imagination of right and left hand movement,” *Electroencephalography and Clinical Neurophysiology*, vol. 103, no. 6, pp. 642–651, dec 1997.

[19] F. Pichiorri, F. De Vico Fallani, F. Cincotti, F. Babiloni, M. Molinari, S. C. Kleih, C. Neuper, A. Kübler, and D. Mattia, “Sensorimotor rhythm-based brain-computer interface training: the impact on motor cortical responsiveness,” *Journal of neural engineering*, vol. 8, no. 2, 2011.

[20] H. Raza, H. Cecotti, and G. Prasad, “Optimising frequency band selection with forward-addition and backward-elimination algorithms in EEG-based brain-computer interfaces,” in *Proceedings of the International Joint Conference on Neural Networks*, vol. 2015-Septe. IEEE, jul 2015, pp. 1–7.

[21] Z. J. Koles, M. S. Lazar, and S. Z. Zhou, “Spatial patterns underlying population differences in the background EEG,” *Brain Topography*, vol. 2, no. 4, pp. 275–284, 1990.

[22] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, “Optimizing spatial filters for robust EEG single-trial analysis,” *IEEE Signal Processing Magazine*, vol. 25, pp. 41–56, 2008.

[23] Repository, https://github.com/bustios/res.

[24] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, “Filter Bank Common Spatial Pattern (FBCSP) algorithm using online adaptive and semi-supervised learning,” in *The 2011 International Joint Conference on Neural Networks*. IEEE, jul 2011, pp. 392–396.

[25] J. F. D. Saa and M. Çetin, “Hidden conditional random fields for classification of imaginary motor tasks from EEG data,” in *Signal Processing Conference, 2011 19th European*. IEEE, 2011, pp. 171–175.

[26] H. Raza, H. Cecotti, and G. Prasad, “A combination of transductive and inductive learning for handling non-stationarities in motor imagery classification,” in *2016 International Joint Conference on Neural Networks (IJCNN)*. IEEE, jul 2016, pp. 763–770.

[27] P. Gaur, R. B. Pachori, and G. Prasad, “An empirical mode decomposition based filtering method for classification of motor-imagery EEG signals for enhancing brain-computer interface,” in *2015 International Joint Conference on Neural Networks (IJCNN)*. IEEE, jul 2015, pp. 1–7.